

HW 5: Theory

Wednesday, October 23, 2024

5:44 AM

1. Gaussian Multivariate

Assume we have a multivariate normal random variable $X = [X_1, X_2, X_3, X_4]^T$, whose covariance matrix Σ and inverse covariance matrix Q are

$$\Sigma = \begin{bmatrix} 0.71 & -0.43 & 0.43 & 0 \\ -0.43 & 0.46 & -0.26 & 0 \\ 0.43 & -0.26 & 0.46 & 0 \\ 0 & 0 & 0 & 0.2 \end{bmatrix} \quad Q = \begin{bmatrix} 5 & 3 & -3 & 0 \\ 3 & 5 & 0 & 0 \\ -3 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Note that Q is simply the inverse of Σ , i.e., $Q = \Sigma^{-1}$.

(a) [5 points] Are X_3 and X_4 correlated?

No, because $\Sigma_{3,4} = 0$.

(b) [5 points] Are X_3 and X_4 conditionally correlated given the other variables? That is, does $\text{cov}(X_3, X_4 | X_1, X_2)$ equal to zero?

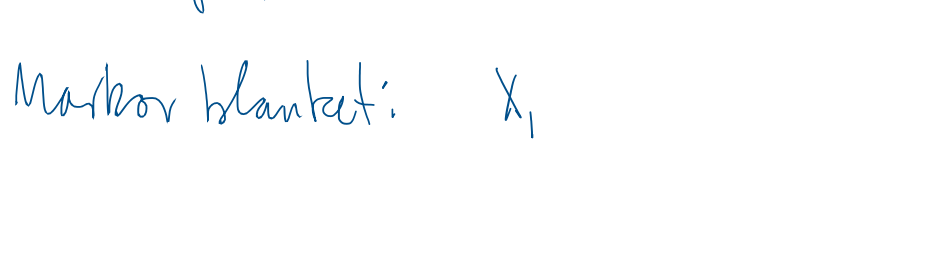
No, X_3 and X_4 are not cond. correlated because $Q_{3,4} = 0$.

The problem should ask if $\text{cov}(X_3, X_4 | X_1, X_2)$ is non-zero.

(c) [5 points] Please find the Markov blanket of X_2 . Recall that the Markov blanket of X_i is the set of variables (denoted by X_{M_i}), such that

$$X_i \perp X_{-\{i\} \cup M_i} | X_{M_i},$$

where $-\{i\} \cup M_i$ denotes all the variables outside of $\{i\} \cup M_i$.



Markov blanket: X_1

(d) [5 points] Assume that $Y = [Y_1, Y_2]^T$ is defined by

$$\begin{aligned} Y_1 &= X_1 + X_4 \\ Y_2 &= X_2 - X_4. \end{aligned}$$

Please calculate the covariance matrix of Y .

$$Y = [Y_1, Y_2]^T$$

$$Y_1 = X_1 + X_4, \quad Y_2 = X_2 - X_4$$

$$\text{Linearity of Cov: } \text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$$

$$\text{Var: } \text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

$$\text{Var}(Y_1) = \text{Var}(X_1 + X_4) = \text{Var}(X_1) + \text{Var}(X_4) + 2 \text{Cov}(X_1, X_4)$$

$$= 0.71 + 0.2 + 0 = 0.91$$

$$\text{Var}(Y_2) = \text{Var}(X_2 - X_4) = \text{Var}(X_2) + \text{Var}(X_4) + 2 \text{Cov}(X_2, X_4)$$

$$= 0.46 + 0.2 + 0 = 0.66$$

$$\text{Cov}(Y_1, Y_2) = \text{Cov}(X_1 + X_4, X_2 - X_4)$$

$$= \text{Cov}(X_1, X_2) + \text{Cov}(X_4, X_2) - \text{Cov}(X_1, X_4) - \text{Var}(X_4)$$

$$= -0.43 + 0 - 0 - 0.2 = -0.63$$

$$\text{Cov}(Y) = \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}$$

2. Expectation Maximization (EM)

Assume we have a dataset of two points $\{x^{(1)}, x^{(2)}\}$:

$$x^{(1)} = -1, \quad x^{(2)} = 1.$$

Assume $x^{(i)}$ is drawn i.i.d. from a simple mixture distribution of two Gaussian components:

$$f(x | \mu_1, \mu_2) = \frac{1}{2} \phi(x | \mu_1, 1) + \frac{1}{2} \phi(x | \mu_2, 1),$$

where $\phi(\cdot | \mu_i, 1)$ denotes the probability density function of Gaussian distribution $\mathcal{N}(\mu_i, 1)$ with mean μ_i and unit variance. We want to estimate the unknown parameters μ_1 and μ_2 .

(a) [5 points] Assume we run EM starting from an initialization of $\mu_1 = -2$ and $\mu_2 = 2$. Please decide the value of μ_1 and μ_2 at the next iteration of EM algorithm. (You may find it handy to know that $1/(1 + \exp(-4)) \approx 0.98$).

$$\gamma_{ik} = P(Z=k | X=x, \theta) = \frac{P(X=x, Z=k | \theta)}{P(X=x | \theta)} = \frac{w_k \times \phi(x | \mu_k, 1)}{\sum_l w_l \times \phi(x | \mu_l, 1)}$$

$$\phi(x | \mu, 1) = (2\pi)^{-1/2} \exp(-\frac{1}{2}(x-\mu)^2)$$

E-Step

$$\begin{aligned} \gamma_{11,1} &= \frac{\frac{1}{2} \phi(-1 | -2, 1)}{\frac{1}{2} \phi(-1 | -2, 1) + \frac{1}{2} \phi(-1 | 2, 1)} \\ &= \frac{\frac{1}{2} (2\pi)^{-1/2} \exp(-\frac{1}{2}(-1-(-2))^2)}{\frac{1}{2} (2\pi)^{-1/2} \exp(-\frac{1}{2}(-1-(-2))^2) + \frac{1}{2} (2\pi)^{-1/2} \exp(-\frac{1}{2}(-1-2)^2)} \\ &= \frac{\exp(-\frac{1}{2}(1))}{\exp(-\frac{1}{2}(1)) + \exp(-\frac{1}{2}(9))} = \frac{1}{1 + \exp(-\frac{3}{2}) / \exp(-\frac{1}{2})} \end{aligned}$$

$$\gamma_{11,1} = \frac{1}{1 + \exp(-4)} = 0.98$$

$$\gamma_{12,1} = 1 - 0.98 = 0.02$$

$$\begin{aligned} \gamma_{22,1} &= \frac{\exp(-\frac{1}{2}(1-2)^2)}{\exp(-\frac{1}{2}(1-(-2))^2) + \exp(-\frac{1}{2}(1-2)^2)} = \frac{\exp(-\frac{1}{2})}{\exp(-\frac{1}{2}(9)) + \exp(-\frac{1}{2})} \\ &= \frac{1}{1 + \exp(-\frac{3}{2})} = 0.98 \quad (\text{could have asserted this result by symmetry}) \end{aligned}$$

$$\gamma_{21,1} = 1 - 0.98 = 0.02$$

M-Step

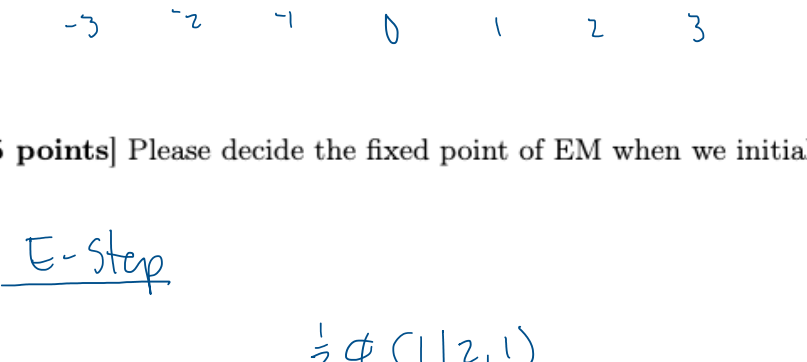
$$\mu_{k,t+1} = \frac{\sum_{i=1}^n \gamma_{ik,t} x^{(i)}}{\sum_{i=1}^n \gamma_{ik,t}}$$

$$\mu_{1,1} = \frac{\gamma_{11,1} x^{(1)} + \gamma_{21,1} x^{(2)}}{\gamma_{11,1} + \gamma_{21,1}} = \frac{0.98(-1) + 0.02(1)}{0.98 + 0.02} = -0.96$$

$$\mu_{2,1} = \frac{\gamma_{12,1} x^{(1)} + \gamma_{22,1} x^{(2)}}{\gamma_{12,1} + \gamma_{22,1}} = \frac{0.02(-1) + 0.98(1)}{0.02 + 0.98} = 0.96$$

(b) [5 points] Do you think EM (when initialized with $\mu_1 = -2$ and $\mu_2 = 2$) will eventually converge to $\mu_1 = -1$ and $\mu_2 = 1$ (i.e., coinciding with the two data points). Please justify your answer using either your theoretical understanding or the result of an empirical simulation.

Intuitively, I'm imagining the marginal plots of these components on a number line. I don't believe there is enough information contained in two symmetric data points to converge on two separate means. It's more likely the algorithm converges to $\mu_1 = \mu_2 = 0$. Imagine the next step of EM with $\mu_1 = -0.96$ and $\mu_2 = 0.96$. Now, there is even more overlap between these two components and the prob. that $x^{(1)}$ came from μ_2 increases relative to the previous step. This of course implies μ_1 and μ_2 will move closer in the M-step. Hence, we see a tendency towards $\mu_1 = \mu_2 = 0$ with each step. A simple simulation would demonstrate this.



(c) [5 points] Please decide the fixed point of EM when we initialize it from $\mu_1 = \mu_2 = 2$.

E-Step

$$\gamma_{21,1} = \frac{\frac{1}{2} \phi(1 | 2, 1)}{\frac{1}{2} \phi(1 | 2, 1) + \frac{1}{2} \phi(1 | 2, 1)} = 0.5$$

$$\gamma_{12,1} = 1 - 0.5 = 0.5$$

$$\gamma_{11,1} = \frac{\frac{1}{2} \phi(-1 | 2, 1)}{\frac{1}{2} \phi(-1 | 2, 1) + \frac{1}{2} \phi(-1 | 2, 1)} = 0.5$$

$$\gamma_{12,1} = 1 - 0.5 = 0.5$$

M-Step

$$\mu_{1,1} = \mu_{2,1} = \frac{0.5(-1) + 0.5(1)}{0.5 + 0.5} = 0$$

Perform another iteration:

E-Step

$$\gamma_{21,2} = \frac{\frac{1}{2} \phi(1 | 2, 1)}{\frac{1}{2} \phi(1 | 2, 1) + \frac{1}{2} \phi(1 | 2, 1)} = \frac{1}{2}$$

$$\Rightarrow \gamma_{21,2} = \gamma_{22,2} = \gamma_{11,2} = \gamma_{12,2} = \frac{1}{2}$$

M-Step

$$\mu_{1,2} = \mu_{2,2} = \frac{0.5(-1) + 0.5(1)}{0.5 + 0.5} = 0$$

μ_1 and μ_2 did not change with another update so $\mu_1 = \mu_2 = 0$ is the fixed point.

(d) [5 points] Please decide the fixed point of **K-means** when we initialize it from $\mu_1 = -2$ and $\mu_2 = 2$.

Assign each point to its nearest cluster:

$$\begin{aligned} \|x^{(1)} - \mu_1\| &= |-1 - (-2)| = 1 \\ \|x^{(1)} - \mu_2\| &= |-1 - 2| = 3 \end{aligned} \quad \left. \begin{aligned} & \\ & \end{aligned} \right\} z^{(1)} = \mu_1 = -2$$

$$\begin{aligned} \|x^{(2)} - \mu_1\|^2 &= |1 - (-2)| = 3 \\ \|x^{(2)} - \mu_2\|^2 &= |1 - 2| = 1 \end{aligned} \quad \left. \begin{aligned} & \\ & \end{aligned} \right\} z^{(2)} = \mu_2 = 2$$

Update centroid:

$$\mu_1 \leftarrow \frac{x^{(1)}}{1} = -1 \quad (\text{mean of points in cluster})$$

$$\mu_2 \leftarrow \frac{x^{(2)}}{1} = 1$$

Since the clusters consist of one point each, the centroids (means) have aligned with their respective point values. In a subsequent step, the points will be assigned to the same clusters and the means cannot move.

Therefore, $(\mu_1, \mu_2) = (-1, 1)$ is the fixed point.